Exercise 16

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Introduction to Electrodynamics

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Solution

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Step 1

We have long coaxial cable that carries a uniform volume charge density ρ on the inner cylinder with radius of a. The cable is electrically neutral, so the outer shell has the same charge density as on the inner cylinder but with negative sign $-\rho$, the radius of the outer shell is b. We need to find the electric field in the three regions $s < a$, $a < s < b$ and $s > b$. First for $s < a$, we consider a cylindrical Gaussian surface with radius of s and length of l, the enclosed charge equals the volume of the cylinder multiplied by the charge density, that is $Q_{\text{enc}} = \rho\pi s^2 l$. Apply Gauss'e law,

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

the electric field is constant throughout the Gauss's surface, so we can pull the electric field out of the integral, and the integration over the surface equals the surface area is $4\pi s^2$, so,

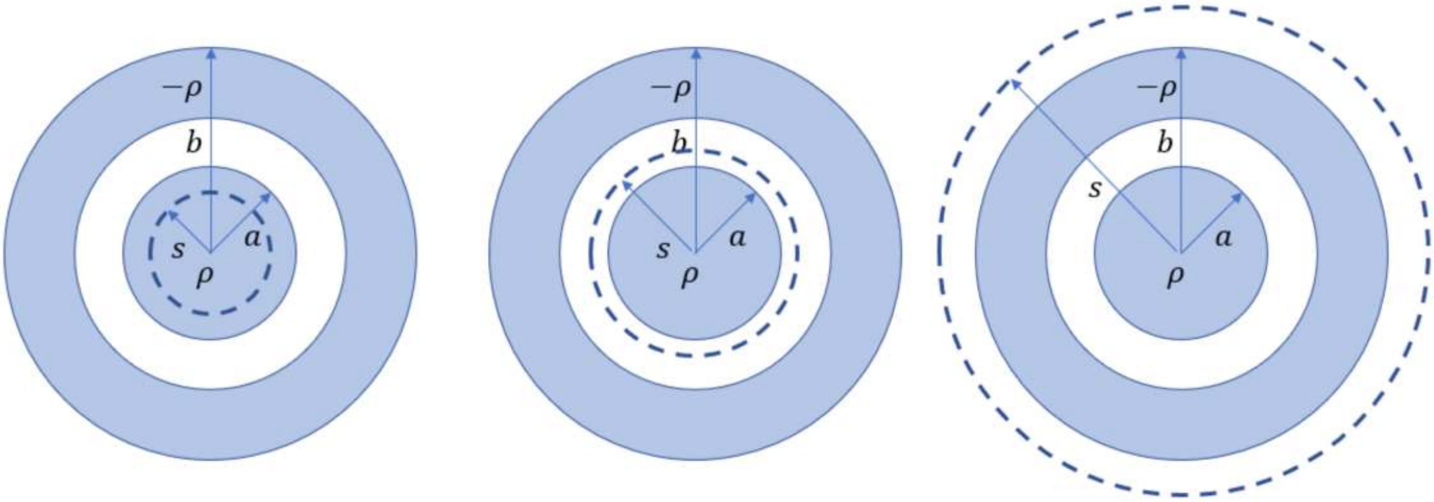
$$E \cdot 4\pi s^2 = \frac{1}{\epsilon_0}\rho\pi s^2 l$$

solve for E we get,

$$\boxed{\mathbf{E}_{s<a} = \frac{\rho s}{2\epsilon_0}\hat{\mathbf{s}}}$$

now we need to find the field within $a < s < b$. The enclosed charge equals the

Step 2

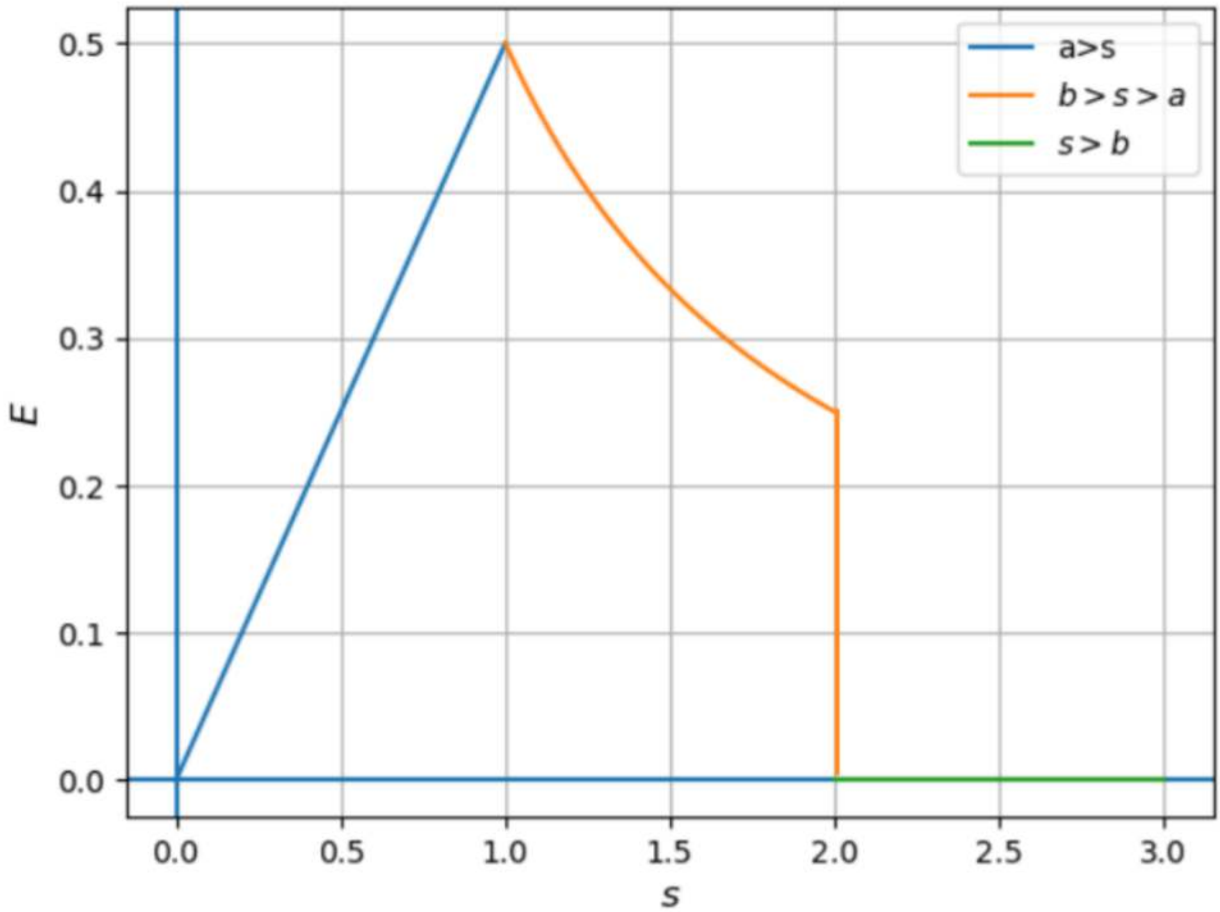


Step 3

Now we need to plot the electric field as a function of s, to do this I used python where the code and the graph are shown in the following figure, for simplicity I let $\rho = \epsilon_0 = 1$.

```
import numpy as np
import matplotlib.pylab as plt
import math as m

a=1
b=2
c=3

s=np.linspace(0,a,100)
s1=np.linspace(a,b,100)
s2=np.linspace(b,c,100)


plt.axhline()
plt.axvline()
plt.grid()
plt.xlabel('$s$ ', fontsize=12)
plt.ylabel('$E$', fontsize=12)

plt.plot(s, s/2,label="a>s")
plt.plot(s1, 1/(2*s1),label="$b>s>a$")

plt.plot(s2, 0*s2,label="$s>b$")

plt.legend()
plt.show()
```

Step 4



Result

$$\mathbf{E}_{s<a} = \frac{\rho s}{2\epsilon_0}\hat{\mathbf{s}}$$

$$\mathbf{E}_{a<s<b} = \frac{\rho a^2}{2\epsilon_0 s}\hat{\mathbf{s}}$$

$$\mathbf{E}_{b<s} = 0$$

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Exercise 17 ›